```
1 7 2
3 5 6
4 8 9
Rectangular
Row Matrix
Enter second matrix:
2 5 6
3 7 8
1 4 3
Not Equal

--------------------------------
Process exited after 46.63 seconds with return value 0
Press any key to continue . . .
```

```c
#include <stdio.h>
int main()
{
    int a[5][5],b[5][5],r,c,i,j;
    int zero=1,diag=1,scalar=1,id=1;
    int sym=1,skew=1,up=1,low=1,eq=1;

    scanf("%d%d",&r,&c);
    for(i=0;i<r;i++)
        for(j=0;j<c;j++) scanf("%d",&a[i][j]);

    for(i=0;i<r;i++)
        for(j=0;j<c;j++){
            if(a[i][j]!=0) zero=0;
            if(i!=j && a[i][j]!=0) diag=scalar=id=0;
            if(i==j && a[i][j]!=a[0][0]) scalar=0;
            if(i==j && a[i][j]!=1) id=0;
            if(i<j && a[i][j]!=0) low=0;
            if(i>j && a[i][j]!=0) up=0;
            if(r==c && a[i][j]!=a[j][i]) sym=0;
            if(r==c && a[i][j]!=-a[j][i]) skew=0;
        }

    if(r==c) printf("Square\n");
    else printf("Rectangular\n");
    if(zero) printf("Zero and Null\n");
```

```c
    if(r==c && low) printf("Lower Triangular\n");
    if(r==c && sym) printf("Symmetric\n");
    if(r==c && skew) printf("Skew-Symmetric\n");
    if(r==1) printf("Row Matrix\n");
    if(c==1) printf("Column Matrix\n");

    printf("Enter second matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++) scanf("%d",&b[i][j]);

    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            if(a[i][j]!=b[i][j]) eq=0;

    if(eq) printf("Equal Matrices\n");
    else printf("Not Equal\n");

    if(r==c){
        if(r==2){
            int d=a[0][0]*a[1][1]-a[0][1]*a[1][0];
            if(d==0) printf("Singular\n");
            else printf("Non-Singular\n");
        } else printf("Check determinant for singularity\n");
    }
    return 0;
}
```